

TELEGRAM_SETUP

Telegram Integration Setup Guide

This guide will walk you through setting up Telegram notifications for HelixCode.

Prerequisites

- Telegram account
- Telegram app (mobile or desktop)

Step 1: Create Telegram Bot

1. Open Telegram and search for @BotFather
2. Start a chat with BotFather
3. Send the command: /newbot
4. BotFather will ask for a name for your bot
 - Enter: HelixCode Notifications
5. BotFather will ask for a username (must end in bot)
 - Enter: helixcode_notifications_bot or similar
6. **Save the bot token** - it looks like:
123456789:ABCdefGHIjklMNOpqrsTUVwxyz

Important: Keep this token secret! It's like a password for your bot.

Step 2: Get Your Chat ID

You need to know where to send notifications. You can send to: - Your personal chat (DM with the bot) - A group chat - A channel

Option A: Using @userinfobot (Easiest)

1. Search for @userinfobot in Telegram
2. Start a chat
3. It will display your chat ID (e.g., 123456789)
4. **Copy this number** - this is your chat ID

Option B: Manual Method

1. Start a chat with your newly created bot (search for its username)

2. Send any message to the bot (e.g., “Hello”)

3. Open this URL in your browser:

`https://api.telegram.org/bot<YOUR_BOT_TOKEN>/getUpdates`

Replace <YOUR_BOT_TOKEN> with your actual bot token

4. Look for "chat":{"id":123456789} in the JSON response

5. **Copy the ID number**

For Group Notifications

1. Create a Telegram group

2. Add your bot to the group (search for it and add as member)

3. Send a message in the group

4. Use the /getUpdates URL method above

5. Group chat IDs start with a minus sign (e.g., -987654321)

Step 3: Configure HelixCode

Option A: Using Environment Variables

Add to your .env file:

```
HELIX_TELEGRAM_BOT_TOKEN=123456789:ABCdefGHIjklMNOpqrsTUVwxyz
HELIX_TELEGRAM_CHAT_ID=123456789
```

Option B: Using config.yaml

Edit config/config.yaml:

```
notifications:
  enabled: true
channels:
  telegram:
    enabled: true
    bot_token: "${HELIX_TELEGRAM_BOT_TOKEN}"
    chat_id: "${HELIX_TELEGRAM_CHAT_ID}"
    timeout: 10
```

Step 4: Test Integration

Using HelixCode CLI (if available)

```
helix notify test --channel telegram --message "Test notification"
```

Using API

```
curl -X POST http://localhost:8080/api/v1/notifications/test \
-H "Content-Type: application/json" \
-H "Authorization: Bearer YOUR_TOKEN" \
-d '{
  "channel": "telegram",
  "title": "Test Notification",
  "message": "Testing Telegram integration from HelixCode",
  "type": "info"
}'
```

Using Go Code

```
import "dev.helix.code/internal/notification"

// Create notification engine
engine := notification.NewNotificationEngine()

// Register Telegram channel
telegramChannel := notification.NewTelegramChannel(
    "123456789:ABCdefGHIjklMNOpqrsTUVwxyz",
    "123456789",
)
engine.RegisterChannel(telegramChannel)

// Send test notification
testNotif := &notification.Notification{
    Title:    "Test Notification",
    Message:  "Testing Telegram integration",
    Type:     notification.NotificationTypeInfo,
    Metadata: map[string]interface{}{
        "task_id": "task-123",
    },
}

err := engine.SendDirect(context.Background(), testNotif,
    []string{"telegram"})
```

Step 5: Configure Notification Rules

Edit `config.yaml` to define when to send Telegram notifications:

```
notifications:
  rules:
    - name: "All failures to Telegram"
      condition: "type==error"
      channels: ["telegram"]
      priority: high
      enabled: true

    - name: "Critical alerts to Telegram"
      condition: "type==alert"
      channels: ["telegram", "slack", "email"]
      priority: urgent
      enabled: true
```

Message Formatting

HelixCode uses HTML formatting for Telegram messages. Messages include:

- **Bold title**
- Message body
- Metadata (if provided) as formatted details

Example output:

****Task Failed****

Task execution failed on worker-123

Details:

- task_id: task-789
- worker_id: worker-123
- error: Connection timeout

Troubleshooting

Error: “Unauthorized”

- **Cause:** Bot token is incorrect
- **Solution:** Verify the bot token from BotFather

Error: “Chat not found”

- **Cause:** Invalid chat ID or you haven’t messaged the bot yet
- **Solution:**
 1. Send a message to your bot first
 2. Verify the chat ID is correct
 3. For groups, make sure the bot is still a member

Error: “Bot was blocked by the user”

- **Cause:** You blocked the bot in Telegram
- **Solution:** Unblock the bot in Telegram settings

No notifications appearing

- **Cause:** Channel might be disabled
- **Solution:** Check `enabled: true` in `config.yaml`

Bot doesn’t respond

- **Cause:** The bot only sends notifications, it doesn’t respond to messages
- **Solution:** This is expected behavior - HelixCode bots are send-only

Advanced Features

HTML Formatting

Telegram supports HTML formatting. You can include:

```
<b>bold text</b>
<i>italic text</i>
<code>monospace code</code>
<pre>code block</pre>
<a href="URL">link text</a>
```

Metadata Display

Automatically includes metadata in notifications:

```
notification := &notification.Notification{
    Title:    "Deployment Complete",
    Message:  "Application deployed successfully",
    Metadata: map[string]interface{}{
        "environment": "production",
        "version":     "v2.1.0",
    },
}
```

```
        "duration": "45s",
    },
}
```

Output:

****Deployment Complete****

Application deployed successfully

Details:

- environment: production
- version: v2.1.0
- duration: 45s

Future: Interactive Buttons

Future versions will support inline keyboards with action buttons: - “Retry Task” - “View Logs” - “Mark as Resolved”

Future: Media Support

Future versions will support: - Sending images (charts, graphs) - Sending documents (log files) - Sending stickers for fun notifications

Security Best Practices

- 1. Never commit bot tokens to version control**
 - Use environment variables
 - Add .env to .gitignore
- 2. Rotate bot tokens if compromised**
 - Use BotFather /revoke command
 - Generate new token with /token
- 3. Use different bots for different environments**
 - Production bot → production chat
 - Staging bot → staging chat
 - Development bot → dev chat
- 4. Limit bot permissions**
 - Bots only need send message permission
 - Don't make bots admins unless necessary

Using with Groups vs Personal Chats

Personal Chat (DM)

- **Pros:** Private, direct notifications
- **Cons:** Only you receive them
- **Use case:** Personal projects, development

Group Chat

- **Pros:** Team visibility, collaboration
- **Cons:** Can get noisy
- **Use case:** Team projects, shared responsibility

Telegram Channel

- **Pros:** Broadcast to many, read-only for subscribers
- **Cons:** One-way communication
- **Use case:** Status updates, announcements

Resources

- [Telegram Bot API Documentation](#)
- [BotFather Commands](#)
- [Telegram Bot Features](#)
- [HelixCode Notification API Documentation](#)

Support

If you encounter issues: 1. Check the [Troubleshooting](#) section 2. Verify bot token and chat ID are correct 3. Check HelixCode logs for error messages 4. Open an issue on GitHub